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Semiconductor packages including mixed bond types and methods of forming same

Abstract

A semiconductor package including hybrid bonding and solder bonding along a first interface and methods of forming the same are disclosed. In an embodiment, a package includes a first interposer, the first interposer including a first redistribution structure; a first die bonded to a first surface of the first redistribution structure with a dielectric-to-dielectric bond and a metal-to-metal bond; a second die bonded to the first surface of the first redistribution structure with a first solder bond; an encapsulant around the first die and the second die; and a plurality of conductive connectors on a second side of the first redistribution structure opposite to the first die and the second die.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8987922	12/2014	Yu et al.	N/A	N/A
8993380	12/2014	Hou et al.	N/A	N/A
9000584	12/2014	Lin et al.	N/A	N/A
9048222	12/2014	Hung et al.	N/A	N/A
9048233	12/2014	Wu et al.	N/A	N/A
9064879	12/2014	Hung et al.	N/A	N/A
9111949	12/2014	Yu et al.	N/A	N/A
9196532	12/2014	Tu et al.	N/A	N/A
9196559	12/2014	Tsai et al.	N/A	N/A
9257333	12/2015	Lu et al.	N/A	N/A
9263511	12/2015	Yu et al.	N/A	N/A
9263839	12/2015	Chen et al.	N/A	N/A
9275924	12/2015	Wang et al.	N/A	N/A
9275925	12/2015	Chen et al.	N/A	N/A
9281254	12/2015	Yu et al.	N/A	N/A
9299649	12/2015	Chiu et al.	N/A	N/A
9368460	12/2015	Yu et al.	N/A	N/A
9372206	12/2015	Wu et al.	N/A	N/A
9425126	12/2015	Kuo et al.	N/A	N/A
9443783	12/2015	Lin et al.	N/A	N/A
9461018	12/2015	Tsai et al.	N/A	N/A
9496189	12/2015	Yu et al.	N/A	N/A

9559081	12/2016	Lai	N/A	H01L 23/5389
9666502	12/2016	Chen et al.	N/A	N/A
9735131	12/2016	Su et al.	N/A	N/A
10535633	12/2019	Wei	N/A	H01L 25/18
11101209	12/2020	Liu et al.	N/A	N/A
11469166	12/2021	Chen et al.	N/A	N/A
11658148	12/2022	Lee et al.	N/A	N/A
11728254	12/2022	Hou et al.	N/A	N/A
2013/0168857	12/2012	Gregorich	257/E23.021	H01L 25/0655
2020/0111729	12/2019	Pan et al.	N/A	N/A
2020/0343218	12/2019	Hu	N/A	H01L 24/80
2021/0143126	12/2020	Choi et al.	N/A	N/A
2022/0262751	12/2021	Zhang et al.	N/A	N/A
2022/0336318	12/2021	Lin	N/A	H01L 21/4882

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
20180114491	12/2017	KR	N/A
20210057853	12/2020	KR	N/A
20210145063	12/2020	KR	N/A
201916304	12/2018	TW	N/A
202119587	12/2020	TW	N/A
2021081855	12/2020	WO	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/362,924, filed on Apr. 13, 2022, which application is hereby incorporated herein by reference.

BACKGROUND

(1) The semiconductor industry has experienced rapid growth due to ongoing improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, and the like). For the most part, improvement in integration density has resulted from iterative reduction of minimum feature size, which allows more components to be integrated into a given area. As the demand for shrinking electronic devices has grown, a need for smaller and more creative packaging techniques of semiconductor dies has emerged.

(2) As semiconductor technologies further advance, stacked and bonded semiconductor devices have emerged as an effective alternative to further reduce the physical size of a semiconductor device. In a stacked semiconductor device, active circuits such as logic, memory, processor circuits and the like are fabricated at least partially on separate substrates and then physically and electrically bonded together in order to form a functional device. Such bonding processes utilize sophisticated techniques, and improvements are desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a cross-sectional view of an integrated circuit die in accordance with some embodiments.

(3) FIGS. 2 through 26 illustrate cross-sectional views of intermediate steps during a process for forming a package component in accordance with some embodiments.

DETAILED DESCRIPTION

(4) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(5) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(6) Various embodiments provide packaged semiconductor devices including mixed bond types at a single interface and methods of forming the same. The method includes hybrid bonding a first die to an interface die and solder bonding a second die to the interface die adjacent the first die. A molding compound may be formed around the first die and the second die, and surfaces of the first die, the second die, and the molding compound may be planarized. In some embodiments, the first die may be a logic die and the second die may be a memory die. In some embodiments, the second die may be a logic die, a passive device die, or a bridge die. The first die and the second die may be electrically coupled to one another through redistribution layers of the interface die. Bonding the first die to the interface die using hybrid bonding allows smaller pitch connections to be formed between the first die and the interface die, reduces the form factor of packages including the first die and the interface die, and improves device performance. Bonding the second die to the interface die using solder bonding reduces costs.

(7) FIG. 1 illustrates a cross-sectional view of an integrated circuit die 50, in accordance with some embodiments. The integrated circuit die 50 will be packaged in subsequent processing to form an integrated circuit package. The integrated circuit die 50 may be a logic die (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a system-on-a-chip (SoC), an application processor (AP), a microcontroller, or the like); a memory die (e.g., a dynamic random access memory (DRAM) die, a static random access memory (SRAM) die, a NAND flash memory die, or the like); a power management die (e.g., a power management integrated circuit (PMIC) die); a radio frequency (RF) die; a sensor die; a micro-electro-mechanical-system (MEMS) die; a signal processing die (e.g., a digital signal processing (DSP) die); a front-end die (e.g., an analog front-end (AFE) die); an input-output (IO) die; a bridge die; the like; or a combination thereof.

(8) The integrated circuit die **50** may be formed in a wafer, which may include different device regions that are singulated in subsequent steps to form a plurality of integrated circuit dies. The integrated circuit die **50** may be processed according to applicable manufacturing processes to form integrated circuits. In some embodiments, the integrated circuit die **50** includes a semiconductor substrate **52**, such as silicon, doped or un-doped, or an active layer of a semiconductor-on-insulator (SOI) substrate. The semiconductor substrate **52** may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may be used. The semiconductor substrate **52** has an active surface (e.g., the surface facing upwards in FIG. 1), sometimes referred to as a front-side, and an inactive surface (e.g., the surface facing downwards in FIG. 1), sometimes referred to as a backside.

(9) Devices (represented by a transistor) **54** may be formed at the front-side of the semiconductor substrate **52**. The devices **54** may be active devices (e.g., transistors, diodes, or the like), capacitors, resistors, or the like. An inter-layer dielectric (ILD) **56** is on the front-side of the semiconductor substrate **52**. The ILD **56** surrounds and may cover the devices **54**. The ILD **56** may include one or more dielectric layers formed of materials such as phosphosilicate glass (PSG), boro-silicate glass (BSG), boron-doped phosphosilicate glass (BPSG), un-doped silicate glass (USG), or the like.

(10) Conductive plugs **58** extend through the ILD **56**, and are electrically and physically coupled to the devices **54**. In embodiments in which the devices **54** are transistors, the conductive plugs **58** may be coupled to gates and source/drain regions (e.g., source region and/or drain regions) of the transistors. The conductive plugs **58** may be formed of tungsten, cobalt, nickel, copper, silver, gold, aluminum, the like, or combinations thereof. An interconnect structure **60** is formed on the ILD **56** and conductive plugs **58**. The interconnect structure **60** interconnects the devices **54** to form integrated circuits. In some embodiments, the interconnect structure **60** may be formed by metallization patterns in dielectric layers on the ILD **56**. The metallization patterns include metal lines and vias formed in one or more low-k dielectric layers. The metallization patterns of the interconnect structure **60** are electrically coupled to the devices **54** by the conductive plugs **58**.

(11) The integrated circuit die **50** further includes pads **62**, such as aluminum pads, to which external connections are made. The pads **62** are on the front-side of the semiconductor substrate **52**, such as in and/or on the interconnect structure **60**. One or more passivation films **64** are on the integrated circuit die **50**, such as on portions of the interconnect structure **60** and the pads **62**.

Openings extend through the passivation films **64** to the pads **62**. Die connectors **66**, such as conductive pillars (e.g., formed of a metal such as copper), extend through the openings in the passivation films **64**. The die connectors **66** may be physically and electrically coupled to respective ones of the pads **62**. The die connectors **66** may be formed by plating, or the like. The die connectors **66** are electrically coupled to the integrated circuits of the integrated circuit die **50**.

(12) Solder regions (e.g., solder balls or solder bumps) may be disposed on the pads **62**. The solder balls may be used to perform chip probe (CP) testing on the integrated circuit die **50**. CP testing may be performed on the integrated circuit die **50** to ascertain whether the integrated circuit die **50** is a known good die (KGD). Thus, only integrated circuit dies **50**, which are KGDs, undergo subsequent processing and are packaged. Dies that fail the CP testing, are not packaged. After testing, the solder regions may be removed in subsequent processing steps.

(13) A dielectric layer **68** may (or may not) be on the front-side of the semiconductor substrate **52**, such as on the passivation films **64** and the die connectors **66**. The dielectric layer **68** laterally encapsulates the die connectors **66**, and the dielectric layer **68** is laterally coterminous with the semiconductor substrate **52**. Initially, the dielectric layer **68** may bury the die connectors **66**, such that the topmost surface of the dielectric layer **68** is above the topmost surfaces of the die connectors **66**. In embodiments in which the solder regions are disposed on the die connectors **66**,

the dielectric layer **68** may bury the solder regions as well. Alternatively, the solder regions may be removed prior to forming the dielectric layer **68**.

(14) The dielectric layer **68** may be a polymer such as PBO, polyimide, BCB, or the like; a nitride such as silicon nitride or the like; an oxide such as silicon oxide, PSG, BSG, BPSG, or the like; the like, or a combination thereof. The dielectric layer **68** may be formed by spin coating, lamination, chemical vapor deposition (CVD), or the like. In some embodiments, the die connectors **66** are exposed through the dielectric layer **68** during formation of the integrated circuit die **50**. In some embodiments, the die connectors **66** remain buried and are exposed during a subsequent process for packaging the integrated circuit die **50**. Exposing the die connectors **66** may remove any solder regions that may be present on the die connectors **66**.

(15) In some embodiments, the integrated circuit die **50** is a stacked device that includes multiple semiconductor substrates **52**. For example, the integrated circuit die **50** may be a memory device, such as a hybrid memory cube (HMC) module, a high bandwidth memory (HBM) module, or the like that includes multiple memory dies. In such embodiments, the integrated circuit die **50** includes multiple semiconductor substrates **52** interconnected by through-substrate vias (TSVs). Each of the semiconductor substrates **52** may (or may not) have an interconnect structure **60**.

(16) FIGS. **2** through **11** illustrate cross-sectional views of intermediate steps during a process for forming a first packaged component **200** (illustrated in FIG. **11**), in accordance with some embodiments. One or more of the integrated circuit dies **50** may be packaged to form the first packaged component **200**. The first packaged component **200** may be referred to as a chip-on-wafer-on-substrate (CoWoS) package or a system on integrated chip (SoIC) package.

(17) In FIG. **2**, an interface die **111** is attached to a carrier substrate **100**. The carrier substrate **100** may be a glass carrier substrate, a ceramic carrier substrate, or the like. The carrier substrate **100** may be a wafer, such that multiple packages can be formed on the carrier substrate **100** simultaneously. Although the interface die **111** is described as a die, the interface die **111** may be a wafer, which may be subsequently singulated.

(18) A release layer **102** is formed on the carrier substrate **100**. The release layer **102** may be formed of a polymer-based material, which may be removed along with the carrier substrate **100** from the interface die **111** in subsequent steps. In some embodiments, the release layer **102** is an epoxy-based thermal-release material, which loses its adhesive property when heated, such as a light-to-heat-conversion (LTHC) release coating. In some embodiments, the release layer **102** may be an ultra-violet (UV) glue, which loses its adhesive property when exposed to UV radiation, such as radiation from UV lights. The release layer **102** may be dispensed as a liquid and cured, may be a laminate film laminated onto the carrier substrate **100**, or the like. A top surface of the release layer **102** may be leveled and may have a high degree of planarity.

(19) The interface die **111** may include an interface substrate **110**, a dielectric layer **112** on the interface substrate **110**, and conductive vias **114** in the dielectric layer **112** and the interface substrate **110**. The interface substrate **110** may be a bulk semiconductor substrate, a semiconductor-on-insulator (SOI) substrate, a multi-layered semiconductor substrate, or the like. The interface substrate **110** may include a semiconductor material, such as silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. The interface substrate **110** may be doped or un-doped. In some embodiments the interface substrate **110** is free from active devices, although the interface substrate **110** may include passive devices formed in and/or on a front surface of the interface substrate **110** (e.g., the surface facing downwards in FIG. **2**), sometimes referred to as a front-side. In embodiments where integrated circuits are formed in the interface substrate **110**, active devices **113** such as transistors, diodes, and

the like, as well as passive devices such as capacitors, resistors, and the like, may be formed in and/or on the front-side of the interface substrate **110**.

(20) The dielectric layer **112** may be formed on the interface substrate **110**. In some embodiments, the dielectric layer **112** is formed of a polymer, which may be a photo-sensitive material such as PBO, polyimide, BCB, or the like, that may be patterned using a lithography mask. In some embodiments, the dielectric layer **112** is formed of a nitride such as silicon nitride; an oxide such as silicon oxide, PSG, BSG, BPSG; or the like. The dielectric layer **112** may be formed by spin coating, lamination, CVD, the like, or a combination thereof.

(21) In some embodiments, the conductive vias **114** may be formed by forming recesses (not separately illustrated) in the dielectric layer **112** and the interface substrate **110**. The recesses may be formed by etching, milling, laser techniques, a combination thereof, or the like. A dielectric material may be formed in the recesses, such as by using an oxidation technique. A barrier layer may be conformally deposited in the openings, such as by CVD, atomic layer deposition (ALD), physical vapor deposition (PVD), thermal oxidation, a combination thereof, or the like. The barrier layer may be formed of an oxide, a nitride, a carbide, a combination thereof, or the like. A conductive material may be deposited over the barrier layer and in the recesses. The conductive material may be formed by an electro-chemical plating process, CVD, ALD, PVD, a combination thereof, or the like. Examples of conductive materials include copper, tungsten, aluminum, silver, gold, a combination thereof, or the like. Excess materials of the conductive material, the barrier layer, and the dielectric material may be removed from surfaces of the dielectric layer **112** using a planarization process. The planarization process may be, for example, a chemical-mechanical polish (CMP), a grinding process, an etch-back process, or the like. Remaining portions of the barrier layer and the conductive material form the conductive vias **114**.

(22) In FIG. **3**, the backside of the interface substrate **110** is thinned. The interface substrate **110** may be thinned by a planarization process applied to the interface substrate **110** to expose the conductive vias **114**. After the conductive vias **114** are exposed, the conductive vias **114** extend through the interface substrate **110** and may be referred to as TSVs. The planarization may remove portions of the interface substrate **110** opposite the dielectric layer **112** such that the conductive vias **114** are exposed. The planarization may be achieved by any suitable process, such as a CMP, a grinding process, an etch-back process, the like, or a combination thereof. After the planarization, the conductive vias **114** may extend completely through the interface substrate **110** and provide interconnection between opposite sides of the interface substrate **110**.

(23) In FIG. **4**, a backside interconnect structure **121** is formed on the interface substrate **110**. The backside interconnect structure **121** includes dielectric layers **116**, metallization layers **118** (also referred to as redistribution layers or redistribution lines) in the dielectric layers **116**, a dielectric layer **120**, and bond pads **122** in the dielectric layer **120**.

(24) The backside interconnect structure **121** may include a plurality of the metallization layers **118** separated from each other by respective layers of the dielectric layers **116**. The metallization layers **118** and the bond pads **122** of the backside interconnect structure **121** are electrically coupled to the conductive vias **114**, and respective ones of the metallization layers **118** may be physically coupled to the conductive vias **114**.

(25) In some embodiments, the dielectric layers **116** are formed of a polymer, which may be a photo-sensitive material such as PBO, polyimide, a BCB-based polymer, or the like, and may be patterned using a lithography mask. In some embodiments, the dielectric layers **116** are formed of a nitride such as silicon nitride; an oxide such as silicon oxide, PSG, BSG, BPSG; or the like. The dielectric layers **116** may be formed by spin coating, lamination, CVD, the like, or a combination thereof. After each dielectric layer **116** is formed, it may be patterned to expose underlying conductive features, such as portions of underlying conductive vias **114** or the underlying metallization layers **118**. The patterning may be by any acceptable process. In embodiments in which the dielectric layers **116** include a photo-sensitive material, the patterning may include

exposing the dielectric layers **116** to light. The dielectric layers **116** may be developed after the exposure. In some embodiments, patterning the dielectric layers **116** may include etching using an anisotropic etch.

(26) The metallization layers **118** each include conductive vias and/or conductive lines. The conductive vias extend through a respective dielectric layer **116**, and the conductive lines extend along the respective dielectric layer **116**, such as on a top surface of the respective dielectric layer **116**. As an example to form a metallization layer **118**, a seed layer (not separately illustrated) is formed over the respective underlying features. For example, the seed layer can be formed on a respective dielectric layer **116**, in openings extending through the respective dielectric layer **116**, and on an underlying feature such as the conductive vias **114** or the metallization layers **118**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using a deposition process, such as PVD, CVD, or the like. A photoresist is formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the metallization layer **118**. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroless plating or electroplating from the seed layer, or the like. The conductive material may comprise a metal or a metal alloy, such as copper, titanium, tungsten, aluminum, the like, or combinations thereof. The photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by any acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed using an acceptable etching process, such as wet or dry etching. The remaining portions of the seed layer and conductive material form the metallization layer **118**.

(27) The dielectric layers **116** and the metallization layers **118** of the backside interconnect structure **121** are illustrated as an example. More or fewer dielectric layers **116** and metallization layers **118** than illustrated may be formed in the backside interconnect structure **121** by repeating or omitting the steps previously described.

(28) The dielectric layer **120** is formed over the dielectric layers **116** and the metallization layers **118**. The dielectric layer **120** may be formed of a material suitable for achieving dielectric-to-dielectric bonds. In some embodiments, the dielectric layer **120** may comprise silicon oxide, silicon nitride, silicon oxynitride, or the like. The dielectric layer **120** may be deposited using a suitable deposition process such as PVD, CVD, ALD, or the like.

(29) The bond pads **122** are formed in the dielectric layer **120**. The bond pads **122** are formed for external connection to the backside interconnect structure **121**. The bond pads **122** may be formed on and extending along top surfaces of the dielectric layers **116**. The bond pads **122** may be physically and electrically coupled to the metallization layers **118**. The bond pads **122** may be electrically coupled to the conductive vias **114** through the metallization layers **118**. The bond pads **122** may be formed of a material and by processes the same as or similar to the metallization layers **118**. In some embodiments, the bond pads **122** have different sizes (e.g., different thicknesses) from the metallization layers **118**. A planarization step may be performed to level surfaces of the bond pads **122** and surfaces of the dielectric layer **120**.

(30) In FIG. 5, a first integrated circuit die **50A** is bonded to the backside interconnect structure **121** by hybrid bonding. A desired type and quantity of integrated circuit dies **50** may be bonded to the backside interconnect structure **121** by hybrid bonding. In the illustrated embodiment, a single first integrated circuit die **50A** is bonded to the backside interconnect structure **121**. The first integrated circuit die **50A** may be a logic device, such as a CPU, a GPU, an SoC, a microcontroller, or the like.

(31) The first integrated circuit die **50A** is bonded to the backside interconnect structure **121** in a hybrid bonding configuration. The first integrated circuit die **50A** are disposed face down such that the front-side of the first integrated circuit die **50A** faces the backside interconnect structure **121** and the backside of the first integrated circuit die **50A** faces away from the backside interconnect structure **121**. This may be referred to as a face-to-back configuration (F2B) as the face of the first integrated circuit die **50A** is toward the back of the interface die **111**.

(32) The dielectric layer **68** of the first integrated circuit die **50A** may be directly bonded to the dielectric layer **120**, and the die connectors **66** of the first integrated circuit die **50A** may be directly bonded to the bond pads **122**. In some embodiments, the bonds between the dielectric layer **68** and the dielectric layer **120** are oxide-to-oxide bonds, or the like. The hybrid bonding process further directly bonds the die connectors **66** of the first integrated circuit die **50A** to the bond pads **122** through direct metal-to-metal bonding. Thus, the first integrated circuit die **50A** is electrically coupled to the backside interconnect structure **121** on the interface die **111** by the physical and electrical connection of the die connectors **66** and the bond pads **122**. In some embodiments, the interface also includes dielectric-to-metal interfaces between the first integrated circuit die **50A** and the backside interconnect structure **121** (e.g., when the die connectors **66** and the bond pads **122** are not perfectly aligned and/or have different widths).

(33) As an example, the hybrid bonding process starts with applying a surface treatment to the dielectric layer **68** and/or the dielectric layer **120**. The surface treatment may include a plasma treatment. The plasma treatment may be performed in a vacuum environment. After the plasma treatment, the surface treatment may further include a cleaning process (e.g., a rinse with deionized water or the like) that may be applied to the dielectric layer **68** and/or the dielectric layer **120**. The die connectors of the first integrated circuit die **50A** may be aligned with the bond pads **122** of the backside interconnect structure **121**. The die connectors **66** may overlap with the corresponding bond pads **122**. A pre-bonding step is performed by placing the first integrated circuit die **50A** in contact with the dielectric layer **120** and the respective bond pads **122** of the backside interconnect structure **121**. The pre-bonding may be performed at room temperature (e.g., between about 21° C. and about 25° C.). After the pre-bonding, an anneal is performed at a temperature between about 150° C. and about 400° C. for a duration between about 0.5 hours and about 3 hours. This causes metal of the die connectors **66** (e.g., copper) and metal of the bond pads **122** (e.g., copper) to inter-diffuse forming direct metal-to-metal bonds.

(34) The first integrated circuit die **50A** is bonded to the backside interconnect structure **121** without the use of solder connections (e.g., micro-bumps or the like). By directly bonding the first integrated circuit die **50A** to the backside interconnect structure **121**, advantages can be achieved, such as, finer bump pitch; small form factor packages by using hybrid bonds; smaller bonding pitch scalability for chip I/O to realize high density die-to-die interconnects; improved mechanical endurance; improved electrical performance; reduced defects; and increased yield. Further, shorter die-to-die distances may be achieved between the first integrated circuit die **50A** and other integrated circuit dies, which has the benefits of smaller form-factor, higher bandwidth, improved power integrity (PI), improved signal integrity (SI), and lower power consumption.

(35) In FIG. 6, bond pads **126** are formed on the bond pads **122** and the dielectric layer **120** of the backside interconnect structure **121** and conductive connectors **128** are formed on the bond pads **126**. The bond pads **126** may be formed by forming a seed layer (not separately illustrated) over the bond pads **122** and the dielectric layer **120**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer including a plurality of sub-layers formed of different materials. In some embodiments, the seed layer includes a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using PVD or the like. A photoresist is formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the bond pads **126**. The patterning forms openings through the photoresist to expose the seed layer. A

conductive material is formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may include a metal, such as copper, titanium, tungsten, aluminum, or the like. The photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed using an acceptable etching process. The remaining portions of the seed layer and the conductive material form the bond pads **126**.

(36) The conductive connectors **128** are formed on the bond pads **126**. The conductive connectors **128** may be ball grid array (BGA) connectors, solder balls, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **128** may be formed of a conductive material that is reflowable, such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **128** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once the layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into desired bump shapes. In some embodiments, the bond pads **126** may be omitted and the conductive connectors **128** may be formed on the bond pads **122**.

(37) In FIG. 7, a second integrated circuit die **50B** is bonded to the backside interconnect structure **121** on the interface die **111** by solder bonding. A desired type and quantity of integrated circuit dies **50** may be bonded to the backside interconnect structure **121** by solder bonding. In the illustrated embodiment, a single second integrated circuit die **50B** is bonded to the backside interconnect structure **121**. The second integrated circuit die **50B** may be a memory device, such as a DRAM die, an SRAM die, a NAND flash die, an HMC module, an HBM module, or the like. Although the second integrated circuit die **50B** is illustrated as a single integrated circuit die, the second integrated circuit die **50B** may include a plurality of stacked integrated circuit dies (also referred to as a die stack).

(38) The second integrated circuit die **50B** is attached to the interface die **111** with solder bonds, such as with the conductive connectors **128**. The second integrated circuit die **50B** may be placed on the backside interconnect structure **121** using, e.g., a pick-and-place tool. Attaching the second integrated circuit die **50B** to the interface die **111** may include placing the second integrated circuit die **50B** on the interface die **111** and reflowing the conductive connectors **128**. The conductive connectors **128** form joints between the bond pads **126** on the interface die **111** and the die connectors **66** of the second integrated circuit die **50B**, electrically coupling the interface die **111** to the second integrated circuit die **50B** through the backside interconnect structure **121**.

(39) An underfill material **129** may be formed around the conductive connectors **128**, and between the second integrated circuit die **50B** and the backside interconnect structure **121**. The underfill material **129** may reduce stress and protect the joints resulting from the reflowing of the conductive connectors **128**. The underfill material **129** may be formed of any suitable underfill material, such as a molding compound, an epoxy, or the like. The underfill material **129** may be formed by a capillary flow process after the second integrated circuit die **50B** is attached to the backside interconnect structure **121**, or may be formed by a suitable deposition method before the second integrated circuit die **50B** is attached to the backside interconnect structure **121**. The underfill material **129** may be applied in a liquid or a semi-liquid form and subsequently cured. In some embodiments, the underfill material **129** is omitted, and the underfill material **129** is omitted in subsequent figures.

(40) The first integrated circuit die **50A** and the second integrated circuit die **50B** may be formed by processes of a same technology node, or may be formed by processes of different technology nodes. For example, the first integrated circuit die **50A** may be of a more advanced process node

than the second integrated circuit die **50B**. The first integrated circuit die **50A** and the second integrated circuit die **50B** may have different sizes (e.g., different heights and/or surface areas), or may have the same size (e.g., the same heights and/or surface areas). Other combinations of the integrated circuit dies **50** are possible. In some embodiments, the first integrated circuit die **50A** and the second integrated circuit die **50B** may have thicknesses greater than about 100 μm .

(41) The first integrated circuit die **50A** and the second integrated circuit die **50B** may be electrically coupled to one another through the interface die **111**. The first integrated circuit die **50A** is physically electrically coupled to the backside interconnect structure **121** through hybrid bonds between the die connectors **66** and the bond pads **122**, and the second integrated circuit die **50B** is physically and electrically coupled to the backside interconnect structure **121** through solder bonds between the die connectors **66** and the bond pads **126**. In some embodiments, the first integrated circuit die **50A** may be a logic die and the second integrated circuit die **50B** may be a memory die. The first integrated circuit die **50A** has a relatively smaller pitch of the die connectors **66** and a higher circuit density, while the second integrated circuit die **50B** has a relatively larger pitch of the die connectors **66** and a lower circuit density. Bonding the first integrated circuit die **50A** to the backside interconnect structure **121** by hybrid bonding achieves advantages, such as finer bump pitch, higher bandwidth, and improved device performance. Bonding the second integrated circuit die **50B** to the backside interconnect structure **121** by solder bonding reduces costs.

(42) In FIG. **8**, an encapsulant **130** is formed on the backside interconnect structure **121** and around the first integrated circuit die **50A**, the second integrated circuit die **50B**, the conductive connectors **128**, the bond pads **126**, and the interface die **111**. After formation, the encapsulant **130** encapsulates the first integrated circuit die **50A**, the second integrated circuit die **50B**, the conductive connectors **128**, the bond pads **126**, the underfill material (if present), and the interface die **111**. The encapsulant **130** may be a molding compound, epoxy, or the like. The encapsulant **130** may be applied by compression molding, transfer molding, or the like. The encapsulant **130** is formed over the interface die **111** such that the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the interface die **111** are buried or covered. The encapsulant **130** is formed in gap regions between the first integrated circuit die **50A** and the second integrated circuit die **50B**. In embodiments in which the underfill material is omitted, the encapsulant **130** may be formed around the conductive connectors **128**, and between the second integrated circuit die **50B** and the backside interconnect structure **121**. The encapsulant **130** may be applied in a liquid or a semi-liquid form and subsequently cured.

(43) The encapsulant **130** is then thinned to expose backsides of the first integrated circuit die **50A** and the second integrated circuit die **50B**. The thinning process may be a grinding process, a CMP, an etch-back, combinations thereof, or the like. After the thinning process, the top surfaces of the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the encapsulant **130** are coplanar (within process variations). The thinning is performed until a desired amount of the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the encapsulant **130** has been removed. Specifically, the thinning removes the portions of the encapsulant **130** covering the top surface of the first integrated circuit die **50A** and the second integrated circuit die **50B** until no encapsulant **130** remains over the first integrated circuit die **50A** and the second integrated circuit die **50B**.

(44) In FIG. **9**, the structure of FIG. **8** is flipped; the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B** are attached to a carrier substrate **140**; and the carrier substrate **100** and the release layer **102** are removed. The device may be flipped such that backsides of the first integrated circuit die **50A** and the second integrated circuit die **50B** face downwards. The carrier substrate **140** may be bonded to the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B** through a release layer **142**. The carrier substrate **140** may be a glass carrier substrate, a ceramic carrier substrate, or the like. The carrier substrate **140** may be a wafer, such that multiple packages can be processed on the carrier substrate

140 simultaneously. The release layer **142** may be formed of a polymer-based material, which may be removed along with the carrier substrate **140** from the overlying structures that will be formed in subsequent steps. In some embodiments, the release layer **142** is an epoxy-based thermal-release material, which loses its adhesive property when heated, such as an LTHC release coating. In some embodiments, the release layer **142** may be a UV glue, which loses its adhesive property when exposed to UV radiation, such radiation from UV lights. The release layer **142** may be dispensed as a liquid and cured, may be a laminate film laminated onto the release layer **142**, or may be the like. The top surface of the release layer **142** may be leveled and may have a high degree of planarity.

(45) A carrier substrate de-bonding is performed to detach (or “de-bond”) the carrier substrate **100** from the interface die **111** and the encapsulant **130**. In some embodiments, the de-bonding includes projecting a light, such as a laser light or an UV light, on the release layer **102** so that the release layer **102** decomposes under the heat of the light and the carrier substrate **100** can be removed. As illustrated in FIG. 9, surfaces of the encapsulant **130**, the conductive vias **114**, and the dielectric layer **112** may be exposed after removing the carrier substrate **100** and the release layer **102**.

(46) In FIG. 10, a front-side interconnect structure **150** is formed on the interface die **111** and the encapsulant **130** opposite the carrier substrate **140**. The front-side interconnect structure **150** includes dielectric layers **152** and metallization layers **154** (sometimes referred to as redistribution layers or redistribution lines) in the dielectric layers **152**. For example, the front-side interconnect structure **150** may include a plurality of metallization layers **154** separated from each other by respective dielectric layers **152**. The metallization layers **154** of the front-side interconnect structure **150** are electrically coupled to the conductive vias **114** of the interface die **111**. The metallization layers **154** are electrically coupled to the first integrated circuit die **50A** and the second integrated circuit die **50B** through the conductive vias **114** and the backside interconnect structure **121**. The first integrated circuit die **50A** and the second integrated circuit die **50B** may be electrically coupled to one another through the metallization layers **118** of the backside interconnect structure **121** and/or the metallization layers **154** of the front-side interconnect structure **150**.

(47) In some embodiments, the dielectric layers **152** are formed of a polymer, which may be a photo-sensitive material such as PBO, polyimide, a BCB-based polymer, or the like, and may be patterned using a lithography mask. In some embodiments, the dielectric layers **152** are formed of a nitride such as silicon nitride; an oxide such as silicon oxide, PSG, BSG, BPSG; or the like. The dielectric layers **152** may be formed by spin coating, lamination, CVD, the like, or a combination thereof. After each dielectric layer **152** is formed, it may be patterned to expose underlying conductive features, such as portions of underlying conductive vias **114** or the underlying metallization layers **154**. The patterning may be by any acceptable process. In embodiments in which the dielectric layers **152** include a photo-sensitive material, the patterning may include exposing the dielectric layers **152** to light. The dielectric layers **152** may be developed after the exposure. In some embodiments, patterning the dielectric layers **152** may include etching using an anisotropic etch.

(48) The metallization layers **154** each include conductive vias and/or conductive lines. The conductive vias extend through a respective dielectric layer **152**, and the conductive lines extend along the respective dielectric layer **152**, such as on a top surface of the respective dielectric layer **152**. As an example to form a metallization layer **154**, a seed layer (not separately illustrated) is formed over the respective underlying features. For example, the seed layer can be formed on a respective dielectric layer **152**, in openings extending through the respective dielectric layer **152**, and on an underlying feature such as the conductive vias **114** or the metallization layers **154**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using a deposition process, such as PVD, CVD, or the like. A photoresist is formed

and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the metallization layer **154**. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroless plating or electroplating from the seed layer, or the like. The conductive material may comprise a metal or a metal alloy, such as copper, titanium, tungsten, aluminum, the like, or combinations thereof. The photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by any acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed using an acceptable etching process, such as wet or dry etching. The remaining portions of the seed layer and conductive material form the metallization layer **154**.

(49) The dielectric layers **152** and the metallization layers **154** of the front-side interconnect structure **150** are illustrated as an example. More or fewer dielectric layers **152** and metallization layers **154** than illustrated may be formed in the front-side interconnect structure **150** by repeating or omitting the steps previously described.

(50) Under-bump metallizations (UBMs) **156** are formed for external connection to the front-side interconnect structure **150**. The UBMs **156** include bump portions on and extending along a top surface of an uppermost dielectric layer of the dielectric layers **152** of the front-side interconnect structure **150**, and include via portions extending through the uppermost dielectric layer of the dielectric layers **152** of the front-side interconnect structure **150**. The via portions may be physically and electrically coupled to an uppermost metallization layer of the metallization layers **154** of the front-side interconnect structure **150**. The UBMs **156** may be electrically coupled to the conductive vias **114**, the first integrated circuit die **50A**, and the second integrated circuit die **50B**. The UBMs **156** may be formed of materials and processes the same as or similar to those used to form the metallization layers **154**. In some embodiments, the UBMs **156** have different sizes (such as greater sizes) than the metallization layers **154**.

(51) Conductive connectors **158** are formed on the UBMs **156**. The conductive connectors **158** may be BGA connectors, solder balls, metal pillars, C4 bumps, micro bumps, ENEPIG-formed bumps, or the like. The conductive connectors **158** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **158** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In some embodiments, the conductive connectors **158** comprise metal pillars (such as copper pillars) formed by a sputtering, printing, electroplating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, a metal cap layer is formed on the top of the metal pillars. The metal cap layer may include nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process.

(52) In FIG. **11**, the carrier substrate **140** is removed and a heat dissipation layer **159** is optionally formed over the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B**. The heat dissipation layer **159**, the encapsulant **130**, the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the interface die **111** form a first packaged component **200**. A carrier substrate de-bonding is performed to detach the carrier substrate **140** from the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B**. In some embodiments, the de-bonding includes projecting a light, such as a laser light or an UV light, on the release layer **142** so that the release layer **142** decomposes under the heat of the light and the carrier substrate **140** can be removed. As illustrated in FIG. **11**, surfaces of the

encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B** may be exposed after removing the carrier substrate **140** and the release layer **142**.

(53) The heat dissipation layer **159** is formed over the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B**. The heat dissipation layer **159** is formed a material with high thermal conductivity, such as a metal or metal nitride. In some embodiments, the heat dissipation layer **159** may be formed of aluminum, titanium, titanium nitride, nickel, nickel vanadium, silver, gold, copper, combinations thereof, or the like. The heat dissipation layer **159** may be conformally formed by a PVD process, such as sputtering or evaporation; a plating process, such as electroless plating or electroplating; a printing process, such as inkjet printing; or the like. In some embodiments, the heat dissipation layer **159** is formed of copper by a sputtering process. The heat dissipation layer **159** may be included to increase heat dissipation from the first integrated circuit die **50A** and the second integrated circuit die **50B**. In some embodiments, the heat dissipation layer **159** may be formed over the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B** before attaching the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B** to the carrier substrate **140**. In some embodiments, the heat dissipation layer **159** may be omitted.

(54) Including both the first integrated circuit die **50A** and the second integrated circuit die **50B** bonded to an interface of the backside interconnect structure **121** by hybrid bonding and solder bonding, respectively, allows for benefits from both hybrid bonding and solder bonding to be achieved. For example, hybrid bonding the first integrated circuit die **50A** allows for dies with small pitches to be bonded to the backside interconnect structure **121**, provides higher bandwidth, and provides improved device performance. Solder bonding the second integrated circuit die **50B** reduces costs, while still providing sufficiently small bond pitches.

(55) FIGS. **2** through **11** illustrate an embodiment in which the package has a face-to-back structure, in which integrated circuit dies **50** have their front surfaces facing the back surface of the interface die **111**. FIGS. **12** through **15** illustrate an embodiment in which a package has a face-to-face structure, in which integrated circuit dies **50** have their front surfaces facing the front surface of the interface die **111**. FIGS. **12** through **15** illustrate cross-sectional views of intermediate steps during a process for forming a second packaged component **300** (illustrated in FIG. **15**), in accordance with some embodiments.

(56) In FIG. **12**, an interface die **111** is provided. The interface die **111** may be the same as or similar to the interface die **111**, discussed above with respect to FIG. **2**. As illustrated in FIG. **12**, the interface die **111** may include an interface substrate **110**, a dielectric layer **112** on a front-side of the interface substrate **110**, active devices **113** formed in and/or on the front-side of the interface substrate **110**, and conductive vias **114** extending through the dielectric layer **112** and into the interface substrate **110**.

(57) In FIG. **13**, a front-side interconnect structure **160** is formed over the interface die **111**. The front-side interconnect structure **160** includes dielectric layers **162**, metallization layers **164** in the dielectric layers **162**, a dielectric layer **166**, and bond pads **168** in the dielectric layer **166**. The front-side interconnect structure **160** may be formed of materials and using processes the same as or similar to those of the backside interconnect structure **121**, described above with respect to FIG. **4**. Specifically, the dielectric layers **162**, the metallization layers **164**, the dielectric layer **166**, and the bond pads **168** may be formed of materials and using processes the same as or similar to the dielectric layers **116**, the metallization layers **118**, the dielectric layer **120**, and the bond pads **122**, respectively.

(58) In FIG. **14**, a first integrated circuit die **50A** and a second integrated circuit die **50B** are bonded to the front-side interconnect structure **160**, an underfill material **129** is optionally formed between the second integrated circuit die **50B** and the front-side interconnect structure **160**, and an encapsulant **130** is formed around the first integrated circuit die **50A** and the second integrated circuit die **50B**. The first integrated circuit die **50A** may be bonded to the front-side interconnect

structure **160** by hybrid bonding, as described above with respect to FIG. 5. The second integrated circuit die **50B** may be bonded to the front-side interconnect structure **160** by solder bonding through bond pads **126** and conductive connectors **128**, as described above with respect to FIGS. 6 and 7. The underfill material **129** may be formed of materials and using processes the same as or similar to the underfill material **129**, described above with respect to FIG. 7. The encapsulant **130** may be formed of materials and using processes the same as or similar to the encapsulant **130**, described above with respect to FIG. 8. Hybrid bonding the first integrated circuit die **50A** allows for dies with small pitches to be bonded to the front-side interconnect structure **160**, provides higher bandwidth, and provides improved device performance. Solder bonding the second integrated circuit die **50B** reduces costs, while still providing sufficiently small bond pitches.

(59) In FIG. 15, the backside of the interface substrate **110** is thinned, a backside interconnect structure **170** is formed on a backside of the interface substrate **110**, and a heat dissipating layer **159** is optionally formed on the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B**. The heat dissipation layer **159**, the encapsulant **130**, the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the interface die **111** form a second packaged component **300**. The interface substrate **110** may be thinned by a planarization process applied to the interface substrate **110** to expose the conductive vias **114**. After the conductive vias **114** are exposed, the conductive vias **114** extend through the interface substrate **110** and may be referred to as TSVs. The planarization may remove portions of the interface substrate **110** opposite the dielectric layer **112** such that the conductive vias **114** are exposed. The planarization may be achieved by any suitable process, such as a CMP, a grinding process, an etch-back process, the like, or a combination thereof. After the planarization, the conductive vias **114** may extend completely through the interface substrate **110** and provide interconnection between opposite sides of the interface substrate **110**.

(60) The backside interconnect structure **170** may be formed of materials and using processes the same as or similar to those of the front-side interconnect structure **150**, described above with respect to FIG. 10. The backside interconnect structure **170** includes dielectric layers **172** and metallization layers **174** in the dielectric layers **172**. The dielectric layers **172** and the metallization layers **174** may be formed of materials and using processes the same as or similar to the dielectric layers **152** and the metallization layers **154**, respectively.

(61) UBMs **176** and conductive connectors **178** are formed for external connection to the backside interconnect structure **170**. The UBMs **176** and the conductive connectors **178** may be formed of materials and using processes the same as or similar to the UBMs **156** and the conductive connectors **158**, respectively, discussed above with respect to FIG. 10. The UBMs **176** include bump portions on and extending along a top surface of an uppermost dielectric layer of the dielectric layers **172** of the backside interconnect structure **170**, and include via portions extending through the uppermost dielectric layer of the dielectric layers **172** of the backside interconnect structure **170**. The via portions may be physically and electrically coupled to an uppermost metallization layer of the metallization layers **174** of the backside interconnect structure **170**. The UBMs **176** may be electrically coupled to the conductive vias **114**, the first integrated circuit die **50A**, and the second integrated circuit die **50B**.

(62) The heat dissipation layer **159** is formed over the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B**. The heat dissipation layer **159** is formed a material with high thermal conductivity, such as a metal or metal nitride. In some embodiments, the heat dissipation layer **159** may be formed of aluminum, titanium, titanium nitride, nickel, nickel vanadium, silver, gold, copper, combinations thereof, or the like. The heat dissipation layer **159** may be conformally formed by a PVD process, such as sputtering or evaporation; a plating process, such as electroless plating or electroplating; a printing process, such as inkjet printing; or the like. In some embodiments, the heat dissipation layer **159** is formed of copper by a sputtering process. The heat dissipation layer **159** may be included to increase heat dissipation from the first integrated

circuit die **50A** and the second integrated circuit die **50B**. In some embodiments, the heat dissipation layer **159** may be formed over the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B** before attaching the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B** to the carrier substrate **140**. In some embodiments, the heat dissipation layer **159** may be omitted.

(63) Including both the first integrated circuit die **50A** and the second integrated circuit die **50B** bonded to an interface of the front-side interconnect structure **160** by hybrid bonding and solder bonding, respectively, allows for benefits from both hybrid bonding and solder bonding to be achieved. For example, hybrid bonding the first integrated circuit die **50A** allows for dies with small pitches to be bonded to the front-side interconnect structure **160**, provides higher bandwidth, and provides improved device performance. Solder bonding the second integrated circuit die **50B** reduces costs, while still providing sufficiently small bond pitches.

(64) FIGS. **16** through **21** illustrate an embodiment in which three integrated circuit dies **50** are bonded to two interface dies **111**. Specifically, as illustrated in FIG. **19**, a first integrated circuit die **50A** is hybrid bonded on a first interface die **111A**, a second integrated circuit die **50B** is solder bonded on the first interface die **111A** and a second interface die **111B**, and a third integrated circuit die **50C** is solder bonded on the second interface die **111B**. FIGS. **16** through **21** illustrate an embodiment in which a package has a face-to-back structure, in which the integrated circuit dies **50** have their front surfaces facing the back surfaces of the interface dies **111**. FIGS. **16** through **21** illustrate cross-sectional views of intermediate steps during a process for forming a third packaged component **400** (illustrated in FIG. **21**), in accordance with some embodiments.

(65) In FIG. **16**, a first interface die **111A** and a second interface die **111B** are attached to a carrier substrate **100**. Front-sides of the first interface die **111A** and the second interface die **111B** are attached to the carrier substrate **100** by a release layer **102**. The carrier substrate **100** and the release layer **102** may be formed of materials and using processes the same as or similar to the carrier substrate **100** and the release layer **102**, respectively, described above with respect to FIG. **2**. The first interface die **111A** and the second interface die **111B** may be the same as or similar to the interface die **111**, described above with respect to FIG. **2**. Each of the first interface die **111A** and the second interface die **111B** may include an interface substrate **110**, a dielectric layer **112** on a front-side of the interface substrate **110**, active devices **113** formed in and/or on the front-side of the interface substrate **110**, and conductive vias **114** extending through the dielectric layer **112** and into the interface substrate **110**. The first interface die **111A** and the second interface die **111B** may be laterally adjacent to one another, and may be separated by a gap.

(66) A backside interconnect structure **121** is formed over a backside of each of the first interface die **111A** and the second interface die **111B**. The backside interconnect structures **121** include dielectric layers **116**, metallization layers **118** in the dielectric layers **116**, dielectric layers **120**, and bond pads **122** in the dielectric layers **120**. The metallization layers **118** are electrically coupled to the conductive vias **114** of the first interface die **111A** and the second interface die **111B**. The bond pads **122** are electrically coupled to the metallization layers **118** and to the conductive vias **114** of the first interface die **111A** and the second interface die **111B** through the metallization layers **118**. The backside interconnect structures **121** may be formed of materials and using processes the same as or similar to those of the backside interconnect structure **121**, described above with respect to FIG. **4**. Specifically, the dielectric layers **116**, the metallization layers **118**, the dielectric layers **120**, and the bond pads **122** may be formed of materials and using processes the same as or similar to the dielectric layers **116**, the metallization layers **118**, the dielectric layer **120**, and the bond pads **122**, respectively.

(67) In FIG. **17**, a first encapsulant **131** is formed on the release layer **102** and around the first interface die **111A** and the second interface die **111B**. The first encapsulant **131** is deposited on the release layer **102** and around the first interface die **111A**, the second interface die **111B**, and the backside interconnect structures **121**. After formation, the first encapsulant **131** encapsulates the

first interface die **111A**, the second interface die **111B**, and the backside interconnect structures **121**. The first encapsulant **131** may be a molding compound, epoxy, or the like. The first encapsulant **131** may be applied by compression molding, transfer molding, or the like. The first encapsulant **131** is formed over the release layer **102** such that the first interface die **111A**, the second interface die **111B**, and the backside interconnect structures **121** are buried or covered. The first encapsulant **131** is formed in gap regions between the first interface die **111A** and the second interface die **111B** and between the backside interconnect structures **121**. The first encapsulant **131** may be applied in a liquid or a semi-liquid form and subsequently cured.

(68) The first encapsulant **131** is then thinned to expose the bond pads **122** of the backside interconnect structures **121**. The thinning process may be a grinding process, a CMP, an etch-back, combinations thereof, or the like. After the thinning process, the top surfaces of the first encapsulant **131**, the bond pads **122**, and the dielectric layers **120** are coplanar (within process variations). The thinning is performed until the bond pads **122** are exposed. In some embodiments, the thinning removes the portions of the first encapsulant **131** covering the backside interconnect structures **121** until none of the first encapsulant **131** remains over the backside interconnect structures **121**.

(69) In FIG. **18**, a first integrated circuit die **50A** is bonded to the backside interconnect structure **121** over the first interface die **111A** by hybrid bonding. A desired type and quantity of integrated circuit dies **50** may be bonded to the backside interconnect structure **121** by hybrid bonding, such as one or more of the first integrated circuit dies **50A**. In the illustrated embodiment, a single first integrated circuit die **50A** is bonded to the backside interconnect structure **121**. The first integrated circuit die **50A** may be a logic device, such as a CPU, a GPU, an SoC, a microcontroller, or the like.

(70) The first integrated circuit die **50A** is bonded to the backside interconnect structure **121** in a hybrid bonding configuration. The first integrated circuit die **50A** is disposed face down such that the front-side of the first integrated circuit die **50A** faces the backside interconnect structure **121** and the backside of the first integrated circuit die **50A** faces away from the backside interconnect structure **121**. This may be referred to as a face-to-back configuration (F2B) as the face of the first integrated circuit die **50A** is toward the back of the first interface die **111A**. The dielectric layer **68** of the first integrated circuit die **50A** may be directly bonded to the dielectric layer **120**, and the die connectors **66** of the first integrated circuit die **50A** may be directly bonded to the bond pads **122**.

(71) The first integrated circuit die **50A** is bonded to the backside interconnect structure **121** without the use of solder connections (e.g., micro-bumps or the like). By directly bonding the first integrated circuit die **50A** to the backside interconnect structure **121**, advantages can be achieved, such as, finer bump pitch; small form factor packages by using hybrid bonds; smaller bonding pitch scalability for chip I/O to realize high density die-to-die interconnects; improved mechanical endurance; improved electrical performance; reduced defects; and increased yield. Further, shorter die-to-die distances may be achieved between the first integrated circuit die **50A** and other integrated circuit dies, which has the benefits of smaller form-factor, higher bandwidth, improved power integrity (PI), improved signal integrity (SI), and lower power consumption.

(72) In FIG. **19**, bond pads **126** are formed on the backside interconnect structures **121** over the first interface die **111A** and the second interface die **111B**, a second integrated circuit die **50B** is bonded to the backside interconnect structures **121** over the first interface die **111A** and the second interface die **111B** by solder bonding, and a third integrated circuit die **50C** is bonded to the backside interconnect structure **121** over the second interface die **111B** by solder bonding. The bond pads **126** and conductive connectors **128** may be formed over the backside interconnect structures **121** of materials and using processes the same as or similar to those of the bond pads **126** and the conductive connectors **128**, respectively, described above with respect to FIG. **6**.

(73) A desired type and quantity of integrated circuit dies **50** may be bonded to the backside interconnect structure **121** by solder bonding, such as one or more of the second integrated circuit

dies **50B** and one or more of the third integrated circuit dies **50C**. In the illustrated embodiment, a single second integrated circuit die **50B** is bonded to the backside interconnect structures **121** over the first interface die **111A** and the second interface die **111B** and a single third integrated circuit die **50C** is bonded to the backside interconnect structure **121** over the second interface die **111B**. The second integrated circuit die **50B** may be a bridge die, which may be a logic device or a passive device. In embodiments in which the second integrated circuit die **50B** is a logic device, the second integrated circuit die **50B** may be a CPU, a GPU, an SoC, a microcontroller, or the like. In embodiments in which the second integrated circuit die **50B** is a passive device, the second integrated circuit die **50B** may be an input-output (IO) die, an SED die, or the like. The third integrated circuit die **50C** may be a memory device, such as a DRAM die, an SRAM die, a NAND flash die, an HMC module, an HBM module, or the like. Although the third integrated circuit die **50C** is illustrated as a single integrated circuit die, the third integrated circuit die **50C** may include a plurality of stacked integrated circuit dies (also referred to as a die stack).

(74) The second integrated circuit die **50B** and the third integrated circuit die **50C** are attached to the backside interconnect structures **121** with solder bonds, such as with the conductive connectors **128**. The second integrated circuit die **50B** and the third integrated circuit die **50C** may be placed on the backside interconnect structures **121** using, e.g., a pick-and-place tool. Attaching the second integrated circuit die **50B** and the third integrated circuit die **50C** to the backside interconnect structures **121** may include placing the second integrated circuit die **50B** and the third integrated circuit die **50C** on the backside interconnect structures **121** and reflowing the conductive connectors **128**. The conductive connectors **128** form joints between the bond pads **126** on the backside interconnect structures **121** and the die connectors **66** of the second integrated circuit die **50B** and the third integrated circuit die **50C**, electrically coupling the first interface die **111A** and the second interface die **111B** to the second integrated circuit die **50B** and the third integrated circuit die **50C** through the backside interconnect structures **121**.

(75) An underfill material **129** may be formed around the conductive connectors **128**, and between the backside interconnect structures **121** and each of the second integrated circuit die **50B** and the third integrated circuit die **50C**. As illustrated in FIG. **19**, the underfill material **129** may be a continuous material, or two separate, discontinuous materials. The underfill material **129** may reduce stress and protect the joints resulting from the reflowing of the conductive connectors **128**. The underfill material **129** may be formed of any suitable underfill material, such as a molding compound, an epoxy, or the like. The underfill material **129** may be formed by a capillary flow process after the second integrated circuit die **50B** and the third integrated circuit die **50C** are attached to the backside interconnect structures **121**, or may be formed by a suitable deposition method before the second integrated circuit die **50B** and the third integrated circuit die **50C** are attached to the backside interconnect structures **121**. The underfill material **129** may be applied in a liquid or a semi-liquid form and subsequently cured. In some embodiments, the underfill material **129** is omitted, and the underfill material **129** is omitted in subsequent figures.

(76) The first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C** may be formed by processes of a same technology node, or may be formed by processes of different technology nodes. For example, the first integrated circuit die **50A** may be of a more advanced process node than the second integrated circuit die **50B** and/or the third integrated circuit die **50C**. The first integrated circuit die **50A**, the second integrated circuit die **50B**, and/or the third integrated circuit die **50C** may have different sizes (e.g., different heights and/or surface areas), or may have the same size (e.g., the same heights and/or surface areas). Other combinations of integrated circuit dies are also possible. In some embodiments, the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C** may have thicknesses greater than about 100 μm .

(77) The first integrated circuit die **50A** and the third integrated circuit die **50C** may be electrically coupled to one another through the backside interconnect structures **121** and the second integrated

circuit die **50B**. The first integrated circuit die **50A** is physically electrically coupled to the backside interconnect structure **121** on the first interface die **111A** through hybrid bonds between the die connectors **66** and the bond pads **122**. The second integrated circuit die **50B** is physically and electrically coupled to the backside interconnect structures **121** on the first interface die **111A** and the second interface die **111B** through solder bonds between the die connectors **66** and the bond pads **126**. The third integrated circuit die **50C** is physically and electrically coupled to the backside interconnect structure **121** on the second interface die **111B** through solder bonds between the die connectors **66** and the bond pads **126**. In some embodiments, the first integrated circuit die **50A** may be a logic die, the second integrated circuit die **50B** may be a bridge die, and the third integrated circuit die **50C** may be a memory die. The first integrated circuit die **50A** has a relatively smaller pitch of the die connectors **66** and a higher circuit density, while the second integrated circuit die **50B** and the third integrated circuit die **50C** have relatively larger pitches of the die connectors **66** and lower circuit densities. Bonding the first integrated circuit die **50A** to the backside interconnect structure **121** by hybrid bonding achieves advantages, such as finer bump pitch, higher bandwidth, and improved device performance. Bonding the second integrated circuit die **50B** and the third integrated circuit die **50C** to the backside interconnect structures **121** by solder bonding reduces costs.

(78) In FIG. **20**, a second encapsulant **133** is formed on the first interface die **111A**, the second interface die **111B**, and the first encapsulant **131** and around the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C**. The second encapsulant **133** may be formed of materials and using processes the same as or similar to the encapsulant **130**, described above with respect to FIG. **8**. The second encapsulant **133** may be thinned to expose backsides of the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C**.

(79) In FIG. **21**, the carrier substrate **100** is removed; a front-side interconnect structure **150** is formed on front-sides of the first interface die **111A** and the second interface die **111B**; and a heat dissipating layer **159** is optionally formed on the second encapsulant **133**, the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C**. The heat dissipation layer **159**, the encapsulant **130**, the first integrated circuit die **50A**, the second integrated circuit die **50B**, the third integrated circuit die **50C**, the first interface die **111A**, and the second interface die **111B** form a third packaged component **400**. A carrier substrate de-bonding is performed to detach the carrier substrate **100** from the first interface die **111A**, the second interface die **111B**, and the first encapsulant **131**. In some embodiments, the de-bonding includes projecting a light, such as a laser light or an UV light, on the release layer **102** so that the release layer **102** decomposes under the heat of the light and the carrier substrate **100** can be removed. As illustrated in FIG. **21**, surfaces of the first interface die **111A**, the second interface die **111B**, and the first encapsulant **131** may be exposed after removing the carrier substrate **100** and the release layer **102**.

(80) The front-side interconnect structure **150** may be formed of materials and using processes the same as or similar to those of the front-side interconnect structure **150**, described above with respect to FIG. **10**. The front-side interconnect structure **150** includes dielectric layers **152** and metallization layers **154** in the dielectric layers **152**. The dielectric layers **152** and the metallization layers **154** may be formed of materials and using processes the same as or similar to the dielectric layers **152** and the metallization layers **154**, respectively.

(81) UBMs **156** and conductive connectors **158** are formed for external connection to the front-side interconnect structure **150**. The UBMs **156** and the conductive connectors **158** may be formed of materials and using processes the same as or similar to the UBMs **156** and the conductive connectors **158**, respectively, discussed above with respect to FIG. **10**. The UBMs **156** include bump portions on and extending along a top surface of an uppermost dielectric layer of the dielectric layers **152** of the front-side interconnect structure **150**, and include via portions extending through the uppermost dielectric layer of the dielectric layers **152** of the front-side interconnect

structure **150**. The via portions may be physically and electrically coupled to an uppermost metallization layer of the metallization layers **154** of the front-side interconnect structure **150**. The UBMs **156** may be electrically coupled to the conductive vias **114**, the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C**.

(82) The heat dissipation layer **159** is formed over the second encapsulant **133**, the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C**. The heat dissipation layer **159** is formed a material with high thermal conductivity, such as a metal or metal nitride. In some embodiments, the heat dissipation layer **159** may be formed of aluminum, titanium, titanium nitride, nickel, nickel vanadium, silver, gold, copper, combinations thereof, or the like. The heat dissipation layer **159** may be conformally formed by a PVD process, such as sputtering or evaporation; a plating process, such as electroless plating or electroplating; a printing process, such as inkjet printing; or the like. In some embodiments, the heat dissipation layer **159** is formed of copper by a sputtering process. The heat dissipation layer **159** may be included to increase heat dissipation from the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C**. In some embodiments, the heat dissipation layer **159** may be formed over the second encapsulant **133**, the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C** before removing the carrier substrate **100**. In some embodiments, the heat dissipation layer **159** may be omitted.

(83) Including both the first integrated circuit die **50A** hybrid bonded to the backside interconnect structure **121** over the first interface die **111A** and the second integrated circuit die **50B** and the third integrated circuit die **50C** solder bonded to the backside interconnect structures **121** over the first interface die **111A** and the second interface die **111B** allows for benefits from both hybrid bonding and solder bonding to be achieved. For example, hybrid bonding the first integrated circuit die **50A** allows for dies with small pitches to be bonded to the backside interconnect structure **121**, provides higher bandwidth, and provides improved device performance. Solder bonding the second integrated circuit die **50B** and the third integrated circuit die **50C** reduces costs, while still providing sufficiently small bond pitches.

(84) FIGS. **22** through **26** illustrate an embodiment in which three integrated circuit dies **50** are bonded to two interface dies **111**. Specifically, as illustrated in FIG. **19**, a first integrated circuit die **50A** is hybrid bonded on a first interface die **111A**, a second integrated circuit die **50B** is solder bonded on the first interface die **111A** and a second interface die **111B**, and a third integrated circuit die **50C** is solder bonded on the second interface die **111B**. FIGS. **22** through **26** illustrate an embodiment in which a package has a face-to-face structure, in which the integrated circuit dies **50** have their front surfaces facing the front surfaces of the interface dies **111**. FIGS. **22** through **26** illustrate cross-sectional views of intermediate steps during a process for forming a fourth packaged component **500** (illustrated in FIG. **26**), in accordance with some embodiments.

(85) In FIG. **22**, a first interface die **111A** and a second interface die **111B** are attached to a carrier substrate **100**. Backsides of the first interface die **111A** and the second interface die **111B** are attached to the carrier substrate **100** by a release layer **102**. The carrier substrate **100** and the release layer **102** may be formed of materials and using processes the same as or similar to the carrier substrate **100** and the release layer **102**, respectively, described above with respect to FIG. **2**. The first interface die **111A** and the second interface die **111B** may be the same as or similar to the interface die **111**, described above with respect to FIG. **2**. Each of the first interface die **111A** and the second interface die **111B** may include an interface substrate **110**, a dielectric layer **112** on a front-side of the interface substrate **110**, active devices **113** formed in and/or on the front-side of the interface substrate **110**, and conductive vias **114** extending through the dielectric layer **112** and into the interface substrate **110**. The first interface die **111A** and the second interface die **111B** may be laterally adjacent to one another, and may be separated by a gap.

(86) A front-side interconnect structure **160** is formed over a front-side of each of the first interface die **111A** and the second interface die **111B**. The front-side interconnect structures **160** include

dielectric layers **162**, metallization layers **164** in the dielectric layers **162**, dielectric layers **166**, and bond pads **168** in the dielectric layers **166**. The metallization layers **164** are electrically coupled to the conductive vias **114** of the first interface die **111A** and the second interface die **111B**. The bond pads **168** are electrically coupled to the metallization layers **164** and to the conductive vias **114** of the first interface die **111A** and the second interface die **111B** through the metallization layers **164**. The front-side interconnect structure **160** may be formed of materials and using processes the same as or similar to those of the backside interconnect structure **121**, described above with respect to FIG. **4**. Specifically, the dielectric layers **162**, the metallization layers **164**, the dielectric layers **166**, and the bond pads **168** may be formed of materials and using processes the same as or similar to the dielectric layers **116**, the metallization layers **118**, the dielectric layer **120**, and the bond pads **122**, respectively.

(87) A first encapsulant **131** is formed on the release layer **102** and around the first interface die **111A** and the second interface die **111B**. The first encapsulant **131** may be formed of materials and using processes the same as or similar to the first encapsulant **131**, discussed above with respect to FIG. **17**. The first encapsulant **131** is thinned to expose the bond pads **168** of the front-side interconnect structure **160**. After the thinning process, the top surfaces of the first encapsulant **131**, the bond pads **168**, and the dielectric layers **166** are coplanar (within process variations). The thinning is performed until the bond pads **168** are exposed. In some embodiments, the thinning removes portions of the first encapsulant **131** covering the front-side interconnect structure **160** until none of the first encapsulant **131** remains over the front-side interconnect structure **160**.

(88) In FIG. **23**, a first integrated circuit die **50A** is bonded to the front-side interconnect structure **160** over the first interface die **111A** by hybrid bonding. A desired type and quantity of integrated circuit dies **50** may be bonded to the front-side interconnect structure **160** by hybrid bonding, such as one or more of the first integrated circuit dies **50A**. In the illustrated embodiment, a single first integrated circuit die **50A** is bonded to the front-side interconnect structure **160**. The first integrated circuit die **50A** may be a logic device, such as a CPU, a GPU, an SoC, a microcontroller, or the like.

(89) The first integrated circuit die **50A** is bonded to the front-side interconnect structure **160** in a hybrid bonding configuration. The first integrated circuit die **50A** is disposed face down such that the front-side of the first integrated circuit die **50A** faces the front-side interconnect structure **160** and the backside of the first integrated circuit die **50A** faces away from the front-side interconnect structure **160**. This may be referred to as a face-to-face configuration (F2F) as the face of the first integrated circuit die **50A** is toward the face of the first interface die **111A**. The dielectric layer **68** of the first integrated circuit die **50A** may be directly bonded to the dielectric layer **166**, and the die connectors **66** of the first integrated circuit die **50A** may be directly bonded to the bond pads **168**.

(90) The first integrated circuit die **50A** is bonded to the front-side interconnect structure **160** without the use of solder connections (e.g., micro-bumps or the like). By directly bonding the first integrated circuit die **50A** to the front-side interconnect structure **160**, advantages can be achieved, such as, finer bump pitch; small form factor packages by using hybrid bonds; smaller bonding pitch scalability for chip I/O to realize high density die-to-die interconnects; improved mechanical endurance; improved electrical performance; reduced defects; and increased yield. Further, shorter die-to-die distances may be achieved between the first integrated circuit die **50A** and other integrated circuit dies, which has the benefits of smaller form-factor, higher bandwidth, improved power integrity (PI), improved signal integrity (SI), and lower power consumption.

(91) In FIG. **24**, bond pads **126** are formed on the front-side interconnect structures **160** over the first interface die **111A** and the second interface die **111B**, a second integrated circuit die **50B** is bonded to the front-side interconnect structures **160** over the first interface die **111A** and the second interface die **111B** by solder bonding, and a third integrated circuit die **50C** is bonded to the front-side interconnect structure **160** over the second interface die **111B** by solder bonding. The bond pads **126** and conductive connectors **128** may be formed over the front-side interconnect structures

160 of materials and using processes the same as or similar to those of the bond pads **126** and the conductive connectors **128**, respectively, described above with respect to FIG. **6**.

(92) A desired type and quantity of integrated circuit dies **50** may be bonded to the front-side interconnect structure **160** by solder bonding, such as one or more of the second integrated circuit dies **50B** and one or more of the third integrated circuit dies **50C**. In the illustrated embodiment, a single second integrated circuit die **50B** is bonded to the front-side interconnect structures **160** over the first interface die **111A** and the second interface die **111B** and a single third integrated circuit die **50C** is bonded to the front-side interconnect structure **160** over the second interface die **111B**. The second integrated circuit die **50B** may be a bridge die, which may be a logic device or a passive device. In embodiments in which the second integrated circuit die **50B** is a logic device, the second integrated circuit die **50B** may be a CPU, a GPU, an SoC, a microcontroller, or the like. In embodiments in which the second integrated circuit die **50B** is a passive device, the second integrated circuit die **50B** may be an input-output (IO) die, an SED die, or the like. The third integrated circuit die **50C** may be a memory device, such as a DRAM die, an SRAM die, a NAND flash die, an HMC module, an HBM module, or the like. Although the third integrated circuit die **50C** is illustrated as a single integrated circuit die, the third integrated circuit die **50C** may include a plurality of stacked integrated circuit dies (also referred to as a die stack).

(93) The second integrated circuit die **50B** and the third integrated circuit die **50C** are attached to the front-side interconnect structures **160** with solder bonds, such as with the conductive connectors **128**. The second integrated circuit die **50B** and the third integrated circuit die **50C** may be placed on the front-side interconnect structures **160** using, e.g., a pick-and-place tool. Attaching the second integrated circuit die **50B** and the third integrated circuit die **50C** to the front-side interconnect structures **160** may include placing the second integrated circuit die **50B** and the third integrated circuit die **50C** on the front-side interconnect structures **160** and reflowing the conductive connectors **128**. The conductive connectors **128** form joints between the bond pads **126** on the front-side interconnect structure **160** and the die connectors **66** of the second integrated circuit die **50B** and the third integrated circuit die **50C**, electrically coupling the first interface die **111A** and the second interface die **111B** to the second integrated circuit die **50B** and the third integrated circuit die **50C** through the front-side interconnect structures **160**.

(94) An underfill material **129** may be formed around the conductive connectors **128**, and between the front-side interconnect structures **160** and each of the second integrated circuit die **50B** and the third integrated circuit die **50C**. As illustrated in FIG. **24**, the underfill material **129** may be a continuous material, or two separate, discontinuous materials. The underfill material **129** may reduce stress and protect the joints resulting from the reflowing of the conductive connectors **128**. The underfill material **129** may be formed of any suitable underfill material, such as a molding compound, an epoxy, or the like. The underfill material **129** may be formed by a capillary flow process after the second integrated circuit die **50B** and the third integrated circuit die **50C** are attached to the front-side interconnect structures **160**, or may be formed by a suitable deposition method before the second integrated circuit die **50B** and the third integrated circuit die **50C** are attached to the front-side interconnect structures **160**. The underfill material **129** may be applied in a liquid or a semi-liquid form and subsequently cured. In some embodiments, the underfill material **129** is omitted, and the underfill material **129** is omitted in subsequent figures.

(95) The first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C** may be formed by processes of a same technology node, or may be formed by processes of different technology nodes. For example, the first integrated circuit die **50A** may be of a more advanced process node than the second integrated circuit die **50B** and/or the third integrated circuit die **50C**. The first integrated circuit die **50A**, the second integrated circuit die **50B**, and/or the third integrated circuit die **50C** may have different sizes (e.g., different heights and/or surface areas), or may have the same size (e.g., the same heights and/or surface areas). Other combinations of integrated circuit dies are also possible. In some embodiments, the first integrated

circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C** may have thicknesses greater than about 100 μm .

(96) The first integrated circuit die **50A** and the third integrated circuit die **50C** may be electrically coupled to one another through the front-side interconnect structures **160** and the second integrated circuit die **50B**. The first integrated circuit die **50A** is physically electrically coupled to the front-side interconnect structure **160** on the first interface die **111A** through hybrid bonds between the die connectors **66** and the bond pads **168**. The second integrated circuit die **50B** is physically and electrically coupled to the front-side interconnect structures **160** on the first interface die **111A** and the second interface die **111B** through solder bonds between the die connectors **66** and the bond pads **126**. The third integrated circuit die **50C** is physically and electrically coupled to the front-side interconnect structure **160** on the second interface die **111B** through solder bonds between the die connectors **66** and the bond pads **126**. In some embodiments, the first integrated circuit die **50A** may be a logic die, the second integrated circuit die **50B** may be a bridge die, and the third integrated circuit die **50C** may be a memory die. The first integrated circuit die **50A** has a relatively smaller pitch of the die connectors **66** and a higher circuit density, while the second integrated circuit die **50B** and the third integrated circuit die **50C** have relatively larger pitches of the die connectors **66** and lower circuit densities. Bonding the first integrated circuit die **50A** to the front-side interconnect structure **160** by hybrid bonding achieves advantages, such as finer bump pitch, higher bandwidth, and improved device performance. Bonding the second integrated circuit die **50B** and the third integrated circuit die **50C** to the front-side interconnect structure **160** by solder bonding reduces costs.

(97) In FIG. **25**, a second encapsulant **133** is formed on the first interface die **111A**, the second interface die **111B**, and the first encapsulant **131** and around the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C**. The second encapsulant **133** may be formed of materials and using processes the same as or similar to the encapsulant **130**, described above with respect to FIG. **8**. The second encapsulant **133** may be thinned to expose backsides of the first integrated circuit die **50A**, the second integrated circuit die **50B**, and the third integrated circuit die **50C**.

(98) In FIG. **26**, the carrier substrate **100** is removed; the backsides of the interface substrates **110** and the first encapsulant **131** are thinned; a backside interconnect structure **170** is formed on the backsides of the interface substrates **110** and the first encapsulant **131**; and a heat dissipating layer **159** is optionally formed on the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B**. The heat dissipation layer **159**, the encapsulant **130**, the first integrated circuit die **50A**, the second integrated circuit die **50B**, the third integrated circuit die **50C**, the first interface die **111A**, and the second interface die **111B** form a fourth packaged component **500**. A carrier substrate de-bonding is performed to detach the carrier substrate **100** from the first interface die **111A**, the second interface die **111B**, and the first encapsulant **131**. In some embodiments, the de-bonding includes projecting a light, such as a laser light or an UV light, on the release layer **102** so that the release layer **102** decomposes under the heat of the light and the carrier substrate **100** can be removed. As illustrated in FIG. **26**, surfaces of the first interface die **111A**, the second interface die **111B**, and the first encapsulant **131** may be exposed after removing the carrier substrate **100** and the release layer **102**.

(99) The interface substrates **110** and the first encapsulant **131** may be thinned by a planarization process applied to the interface substrates **110** and the first encapsulant **131** to expose the conductive vias **114**. After the conductive vias **114** are exposed, the conductive vias **114** extend through the interface substrates **110** and may be referred to as TSVs. The planarization may remove portions of the interface substrates **110** opposite the dielectric layer **112** such that the conductive vias **114** are exposed. The planarization may be achieved by any suitable process, such as a CMP, a grinding process, an etch-back process, the like, or a combination thereof. After the planarization, the conductive vias **114** may extend completely through the interface substrates **110** and provide

interconnection between opposite sides of the interface substrates **110**.

(100) The backside interconnect structure **170** may be formed of materials and using processes the same as or similar to those of the front-side interconnect structure **150**, described above with respect to FIG. **10**. The backside interconnect structure **170** includes dielectric layers **172** and metallization layers **174** in the dielectric layers **172**. The dielectric layers **172** and the metallization layers **174** may be formed of materials and using processes the same as or similar to the dielectric layers **152** and the metallization layers **154**, respectively.

(101) UBMs **176** and conductive connectors **178** are formed for external connection to the backside interconnect structure **170**. The UBMs **176** and the conductive connectors **178** may be formed of materials and using processes the same as or similar to the UBMs **156** and the conductive connectors **158**, respectively, discussed above with respect to FIG. **10**. The UBMs **176** include bump portions on and extending along a top surface of an uppermost dielectric layer of the dielectric layers **172** of the backside interconnect structure **170**, and include via portions extending through the uppermost dielectric layer of the dielectric layers **172** of the backside interconnect structure **170**. The via portions may be physically and electrically coupled to an uppermost metallization layer of the metallization layers **174** of the backside interconnect structure **170**. The UBMs **176** may be electrically coupled to the conductive vias **114**, the first integrated circuit die **50A**, and the second integrated circuit die **50B**.

(102) The heat dissipation layer **159** is formed over the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B**. The heat dissipation layer **159** is formed a material with high thermal conductivity, such as a metal or metal nitride. In some embodiments, the heat dissipation layer **159** may be formed of aluminum, titanium, titanium nitride, nickel, nickel vanadium, silver, gold, copper, combinations thereof, or the like. The heat dissipation layer **159** may be conformally formed by a PVD process, such as sputtering or evaporation; a plating process, such as electroless plating or electroplating; a printing process, such as inkjet printing; or the like. In some embodiments, the heat dissipation layer **159** is formed of copper by a sputtering process. The heat dissipation layer **159** may be included to increase heat dissipation from the first integrated circuit die **50A** and the second integrated circuit die **50B**. In some embodiments, the heat dissipation layer **159** may be formed over the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B** before attaching the encapsulant **130**, the first integrated circuit die **50A**, and the second integrated circuit die **50B** to the carrier substrate **140**. In some embodiments, the heat dissipation layer **159** may be omitted.

(103) Including both the first integrated circuit die **50A** hybrid bonded to the front-side interconnect structure **160** over the first interface die **111A** and the second integrated circuit die **50B** and the third integrated circuit die **50C** solder bonded to the front-side interconnect structures **160** over the first interface die **111A** and the second interface die **111B** allows for benefits from both hybrid bonding and solder bonding to be achieved. For example, hybrid bonding the first integrated circuit die **50A** allows for dies with small pitches to be bonded to the front-side interconnect structure **160**, provides higher bandwidth, and provides improved device performance. Solder bonding the second integrated circuit die **50B** and the third integrated circuit die **50C** reduces costs, while still providing sufficiently small bond pitches.

(104) Embodiments may achieve advantages. Bonding the first integrated circuit dies **50A** to the interface dies **111** through hybrid bonding allows for dies with small pitches to be bonded to the interface dies **111**, provides higher bandwidth, and provides improved device performance. Bonding the second integrated circuit dies **50B** and/or the third integrated circuit dies **50C** to the interface dies **111** through solder bonding reduces costs, while still providing sufficiently small bond pitches.

(105) The bonding processes described above have been described as being at the die level, wherein the integrated circuit dies **50** are bonded to the interface dies **111** after the integrated circuit dies **50** are bonded to the interface dies **111** are singulated into separate dies. Alternatively, the

bonding may be performed at the die-to-wafer level, or the wafer-to-wafer level, and subsequent singulation processes may be performed.

(106) In accordance with an embodiment, a package includes a first interposer, the first interposer including a first redistribution structure; a first die bonded to a first surface of the first redistribution structure with a dielectric-to-dielectric bond and a metal-to-metal bond; a second die bonded to the first surface of the first redistribution structure with a first solder bond; an encapsulant around the first die and the second die; and a plurality of conductive connectors on a second side of the first redistribution structure opposite to the first die and the second die. In an embodiment, the first die includes a logic die, and the second die includes a memory die. In an embodiment, the package further includes a second interposer adjacent the first interposer, the second interposer including a second redistribution structure, the second die being bonded to a first surface of the second redistribution structure with a second solder bond. In an embodiment, the package further includes a third die bonded to the first surface of the second redistribution structure with a third solder bond. In an embodiment, the package further includes a second encapsulant extending from the first interposer to the second interposer. In an embodiment, a top surface of the encapsulant, a top surface of the first die, and a top surface of the second die are level with one another. In an embodiment, the package further includes an under-bump metallization on the first surface of the first redistribution structure, the second die being bonded to the first redistribution structure by the first solder bond on the under-bump metallization, and a surface of the under-bump metallization is level with a surface of the first die.

(107) In accordance with another embodiment, a semiconductor package includes a first interface die; a second interface die adjacent the first interface die; a first encapsulant extending from the first interface die to the second interface die; a first integrated circuit die bonded to the first interface die by a dielectric-to-dielectric bond and a metal-to-metal bond; and a second integrated circuit die bonded to the second interface die by a first solder bond. In an embodiment, the second integrated circuit die is further bonded to the first interface die by a second solder bond. In an embodiment, the semiconductor package further includes a third integrated circuit die bonded to the second interface die by a third solder bond. In an embodiment, the first integrated circuit die is a logic die, the second integrated circuit die is a bridge die, and the third integrated circuit die is a memory die. In an embodiment, the first interface die includes a first interconnect structure, the first integrated circuit die is bonded to a first surface of the first interconnect structure, the second interface die includes a second interconnect structure, the second integrated circuit die is bonded to a second surface of the second interconnect structure, and the first surface and the second surface are level. In an embodiment, the semiconductor package further includes a second encapsulant around the first integrated circuit die and the second integrated circuit die, the second encapsulant physically contacting the first integrated circuit die, the second integrated circuit die, the first interface die, and the second interface die.

(108) In accordance with yet another embodiment, a method includes providing a first interposer including a first interconnect structure on a first interposer substrate; bonding a first die to the first interconnect structure, bonding the first die including directly bonding a first insulating layer of the first die to a bonding layer of the first interconnect structure and directly bonding a first die connector of the first die to a first bond pad of the first interconnect structure; bonding a second die to the first interconnect structure, bonding the second die including solder bonding a second die connector of the second die to a second bond pad of the first interconnect structure; and encapsulating the first die and the second die in a molding compound. In an embodiment, the method further includes forming a first under-bump metallization on the second bond pad; forming a conductive connector on the first under-bump metallization; and reflowing the conductive connector to bond the second die to the first interconnect structure. In an embodiment, the method further includes planarizing the molding compound, the first die, and the second die. In an embodiment, the method further includes forming an underfill material between the second die and

the first interconnect structure, the underfill material surrounding solder joints formed between the second die and the first interconnect structure. In an embodiment, the molding compound is formed surrounding solder joints formed between the second die and the first interconnect structure. In an embodiment, the method further includes providing a second interposer adjacent the first interposer, the second interposer including a second interconnect structure on a second interposer substrate, bonding the second die to the first interconnect structure bonds the second die to the second interconnect structure by solder bonding. In an embodiment, the method further includes bonding a third die to the second interconnect structure, bonding the third die includes solder bonding a third die connector of the third die to a third bond pad of the second interconnect structure.

(109) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A package comprising: a first interposer, wherein the first interposer comprises a first redistribution structure; a first die bonded to a first surface of the first redistribution structure with a dielectric-to-dielectric bond and a metal-to-metal bond; a second die bonded to the first surface of the first redistribution structure with a first solder bond; an encapsulant around the first die and the second die; and a plurality of conductive connectors on a second side of the first redistribution structure opposite to the first die and the second die.
2. The package of claim 1, wherein the first die comprises a logic die, and wherein the second die comprises a memory die.
3. The package of claim 1, further comprising a second interposer adjacent the first interposer, wherein the second interposer comprises a second redistribution structure, wherein the second die is bonded to a first surface of the second redistribution structure with a second solder bond.
4. The package of claim 3, further comprising a third die bonded to the first surface of the second redistribution structure with a third solder bond.
5. The package of claim 3, further comprising a second encapsulant extending from the first interposer to the second interposer.
6. The package of claim 1, wherein a top surface of the encapsulant, a top surface of the first die, and a top surface of the second die are level with one another.
7. The package of claim 1, further comprising an under-bump metallization on the first surface of the first redistribution structure, wherein the second die is bonded to the first redistribution structure by the first solder bond on the under-bump metallization, and wherein a surface of the under-bump metallization is level with a surface of the first die.
8. A semiconductor package comprising: a first interface die; a second interface die adjacent the first interface die; a first encapsulant extending from the first interface die to the second interface die; a first integrated circuit die bonded to the first interface die by a dielectric-to-dielectric bond and a metal-to-metal bond; and a second integrated circuit die bonded to the second interface die by a first solder bond.
9. The semiconductor package of claim 8, wherein the second integrated circuit die is further bonded to the first interface die by a second solder bond.
10. The semiconductor package of claim 9, further comprising a third integrated circuit die bonded

to the second interface die by a third solder bond.

11. The semiconductor package of claim 10, wherein the first integrated circuit die is a logic die, wherein the second integrated circuit die is a bridge die, and wherein the third integrated circuit die is a memory die.

12. The semiconductor package of claim 8, wherein the first interface die comprises a first interconnect structure, wherein the first integrated circuit die is bonded to a first surface of the first interconnect structure, wherein the second interface die comprises a second interconnect structure, wherein the second integrated circuit die is bonded to a second surface of the second interconnect structure, and wherein the first surface and the second surface are level.

13. The semiconductor package of claim 8, further comprising a second encapsulant around the first integrated circuit die and the second integrated circuit die, wherein the second encapsulant physically contacts the first integrated circuit die, the second integrated circuit die, the first interface die, and the second interface die.

14. A method comprising: providing a first interposer comprising a first interconnect structure on a first interposer substrate; bonding a first die to the first interconnect structure, wherein bonding the first die comprises directly bonding a first insulating layer of the first die to a bonding layer of the first interconnect structure and directly bonding a first die connector of the first die to a first bond pad of the first interconnect structure; bonding a second die to the first interconnect structure, wherein bonding the second die comprises solder bonding a second die connector of the second die to a second bond pad of the first interconnect structure; and encapsulating the first die and the second die in a molding compound.

15. The method of claim 14, further comprising: forming a first under-bump metallization on the second bond pad; forming a conductive connector on the first under-bump metallization; and reflowing the conductive connector to bond the second die to the first interconnect structure.

16. The method of claim 14, further comprising planarizing the molding compound, the first die, and the second die.

17. The method of claim 14, further comprising forming an underfill material between the second die and the first interconnect structure, the underfill material surrounding solder joints formed between the second die and the first interconnect structure.

18. The method of claim 14, wherein the molding compound is formed surrounding solder joints formed between the second die and the first interconnect structure.

19. The method of claim 14, further comprising providing a second interposer adjacent the first interposer, the second interposer comprising a second interconnect structure on a second interposer substrate, wherein bonding the second die to the first interconnect structure bonds the second die to the second interconnect structure by solder bonding.

20. The method of claim 19, further comprising bonding a third die to the second interconnect structure, wherein bonding the third die comprises solder bonding a third die connector of the third die to a third bond pad of the second interconnect structure.
